



Team MACS: TaskBoard Quality Assurance & Automation Strategy

Team MACS - Maurice, Anthony, Cory, and Samuel.

Maurice Grant (Introduction & Test Overview)

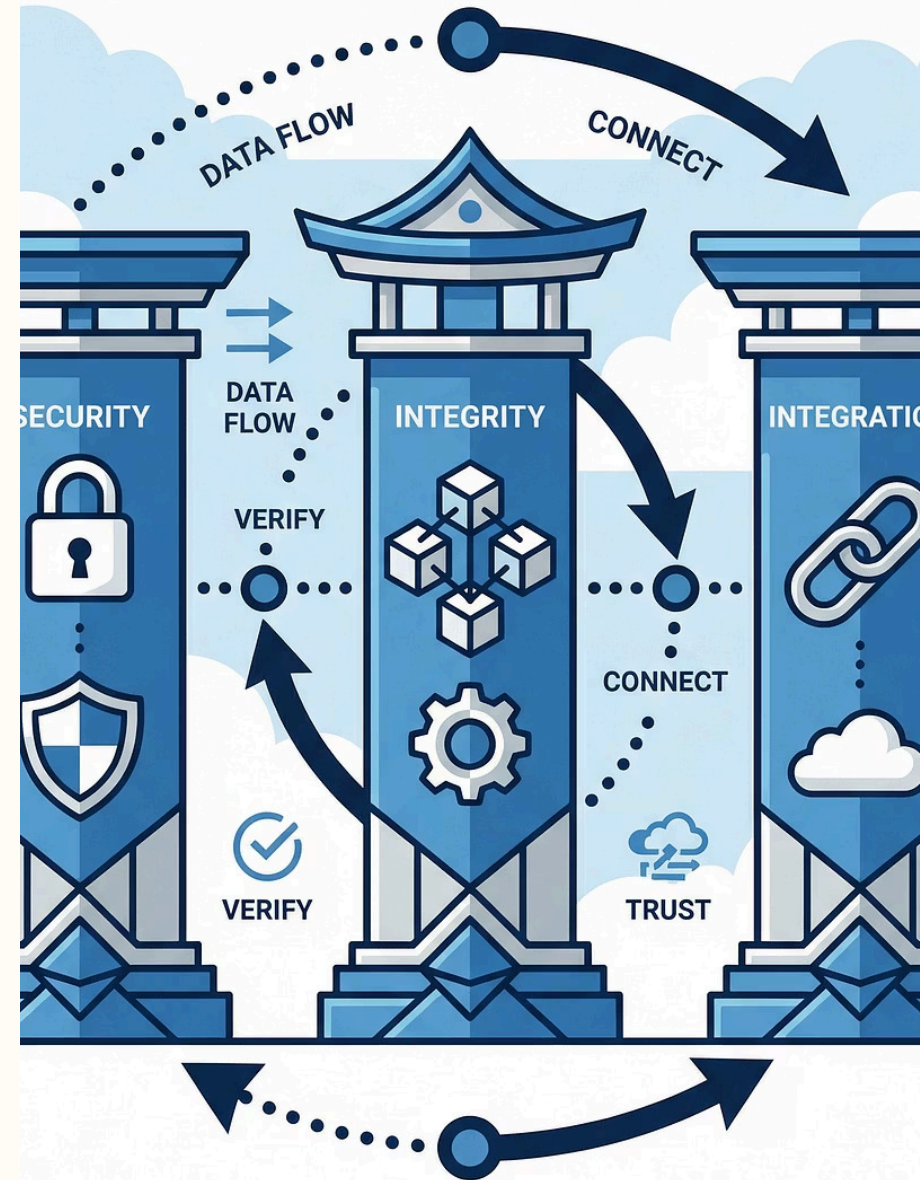
Project Recap & Testing Scope

TaskBoard Overview

JIRA-style issue tracking with role-based permissions (Admin, Tester, Developer) managing the full issue lifecycle across Projects.

Our Testing Pillars

- **Security:** Validating role-specific access controls and JWT authentication mechanisms
- **Integrity:** Ensuring consistent database states through comprehensive CRUD operations
- **Integration:** Verifying seamless communication between Spring Boot API and Angular frontend



Testing Framework & Tools Architecture



Backend Validation

RestAssured Framework

Provides fast, isolated testing of business logic at the API level. Ensures JSON schema integrity and validates proper HTTP status code responses across all endpoints.



BDD Strategy

Cucumber & Gherkin

Uses Given/When/Then syntax to translate business requirements into executable test scripts, creating living documentation that stakeholders can understand.

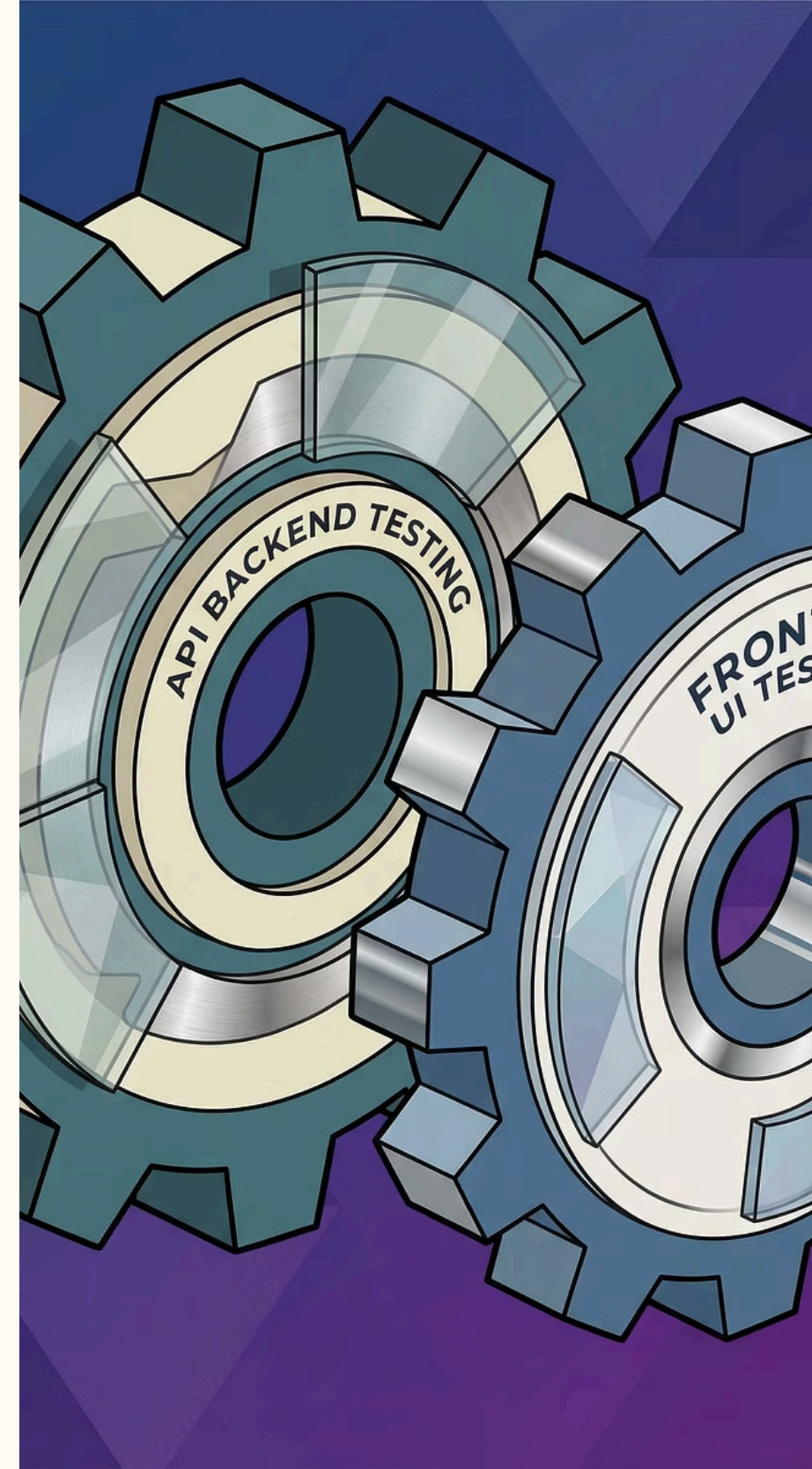


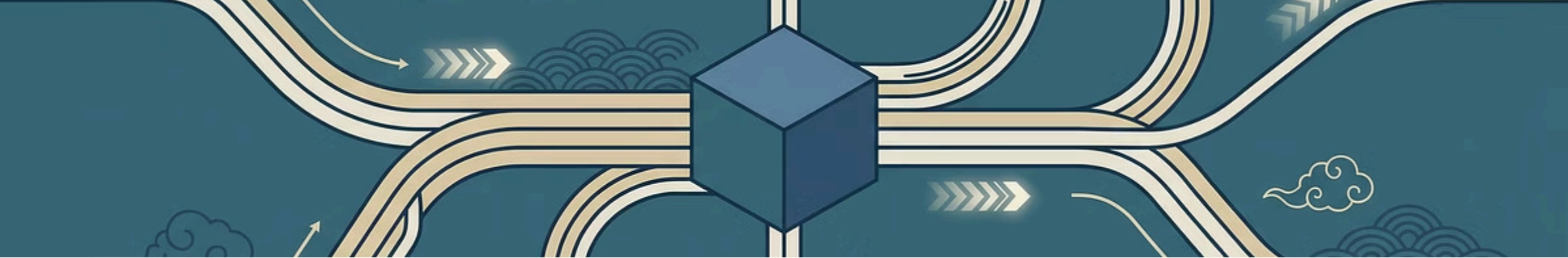
E2E Validation

Selenium WebDriver

Simulates real user behavior in the browser to verify critical UI workflows, ensuring the frontend accurately reflects backend state changes.

We believed this combined API-level precision with browser-based user simulation, created the best testing strategy that catches issues across the architectural layer.





High-Level Automation Overview

01

Full Spectrum Coverage

Automated validation spans Project management workflows, Issue reporting systems, and collaborative commenting features ensuring complete business logic verification.

03

Parallel Execution

Running backend API tests and frontend UI tests concurrently accelerates feedback cycles and helps pinpoint whether failures originate from architectural issues or integration problems.

02

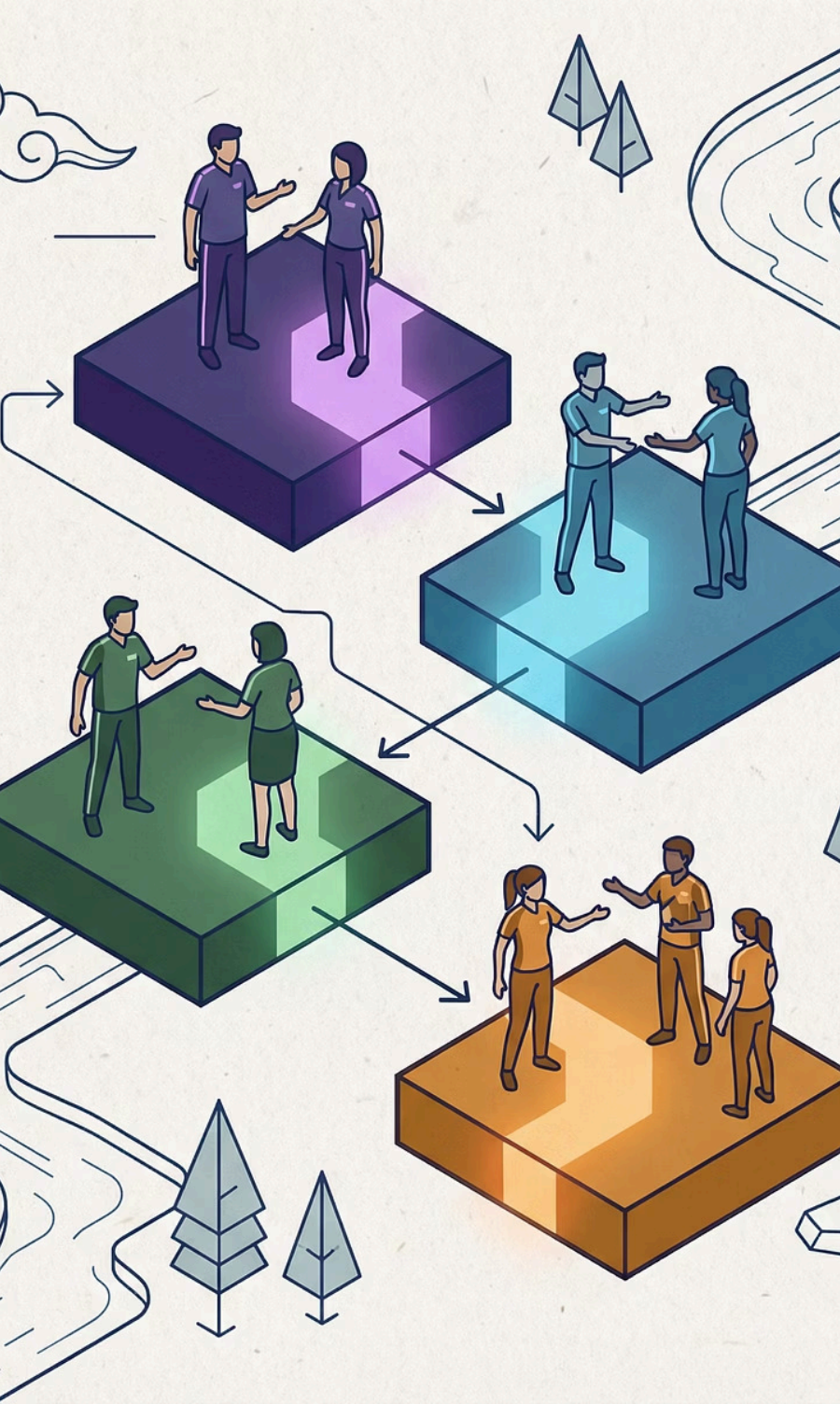
Data Isolation Strategy

Leveraging dynamic environment setups with isolated test databases ensures tests remain repeatable, independent, and free from cross-contamination between test runs.

04

Push-Button Verification

Our success metric: a reliable, automated validation process that verifies every build against core business requirements without manual intervention, significantly reducing regression risk.



TEAM PRESENTATION

Presentation Division Strategy

Team MACS has structured this presentation to provide comprehensive coverage of our automation framework, from strategic vision through technical implementation to retrospective insights.



Maurice: Introduction & Testing Overview

Establishing our testing philosophy, framework selection rationale, and high-level automation scope .



Cory: Test Execution & Reporting

Live demonstration of automated test suites in action, plus walkthrough of test report.



Anthony: DevOps Integration

Docker containerization strategy ensuring consistent development environments and enabling true "one-click" setup across all team members.



Samuel: Retrospective & Summary

Analysis of what went well, areas for improvement, and a summary of our key technical accomplishments